Tuan Duc **Ngo**

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Education

University of Massachusetts Amherst — *Amherst, MA*

PH.D. IN COMPUTER SCIENCE — *Sep 2023 – Present*

- Advisors: Prof. Evangelos Kalogerakis, Prof. Chuang Gan
- GPA: 3.97 / 4.00

Ho Chi Minh City University of Technology - VNU — *Ho Chi Minh City, Vietnam*

B.E. IN COMPUTER ENGINEERING — *Aug 2017 – Aug 2021*

- Graduated with the *Highest Honor*, Rank: 1/400
- GPA: 9.62 / 10.00

Research Experience

Meta Reality Labs — *Redmond, WA*

RESEARCH INTERN, XR-CORE AI TEAM — *Jun 2026 – present*

Adobe Inc. — *San Jose, CA*

RESEARCH INTERN, VIDEO TEAM — *May 2025 – Nov 2025*

- Worked with Gabriel Huang, Dr. Joon-Young Lee, Dr. Seoung Wug Oh
- Worked on 3D reconstruction
- Published one paper at CVPR 2026.

Snap Inc. — *Santa Monica, CA*

RESEARCH INTERN, CREATIVE VISION TEAM — *May 2024 – May 2025*

- Worked with Dr. Chaoyang Wang, Dr. Hsin-Ying Lee, Dr. Peiye Zhuang
- Worked on 4D reconstruction and 4D generative model
- Published two papers, one at ICLR 2025 and one at CVPR 2025.

Qualcomm AI Research *(formerly VinAI Research)* — *Ho Chi Minh City, Vietnam*

RESEARCH RESIDENT, VISION TEAM — *Aug 2021 – Jul 2023*

- Worked with Dr. Khoi Nguyen and Prof. Binh-Son Hua
- Worked on 2D/3D understanding
- Published three papers, one at ECCV 2022, one at CVPR 2023, and one at ICCV 2023.

Publications

CONFERENCES

1. **VolFill: Single-View Amodal 3D Scene Reconstruction with Volumetric Flow Matching** [Paper] [Webpage]
   Tuan Duc Ngo, Chuang Gan, Evangelos Kalogerakis
   NeurIPS 2026

2. **DAGE: Dual-Stream Architecture for Efficient and Fine-Grained Geometry Estimation** [Paper] [Webpage]
   Tuan Duc Ngo, Jiahui Huang, Seoung Wug Oh, Kevin Blackburn-Matzen, Evangelos Kalogerakis, Chuang Gan, Joon-Young Lee
   CVPR 2026

3. **4Real-Video: Learning Generalizable Photo-Realistic 4D Video Diffusion** [Paper] [Webpage]
   Chaoyang Wang*, Peiye Zhuang*, **Tuan Duc Ngo***, Willi Menapace, Aliaksandr Siarohin, Michael Vasilkovsky, Ivan Skorokhodov, Sergey Tulyakov, Peter Wonka, Hsin-Ying Lee
   CVPR 2025 **Highlight**

4. **DELTA: Dense Efficient Long-range 3D Tracking for any video** [Paper] [Webpage]
   Tuan Duc Ngo, Peiye Zhuang, Chuang Gan, Evangelos Kalogerakis, Sergey Tulyakov, Hsin-Ying Lee, Chaoyang Wang
   ICLR 2025

5. **Open3DIS: Open-vocabulary 3D Instance Segmentation with 2D Mask Guidance** [Paper] [Webpage]
   Phuc Nguyen*, **Tuan Duc Ngo***, Chuang Gan, Evangelos Kalogerakis, Anh Tran, Cuong Pham, Khoi Nguyen
   CVPR 2024

6. **GaPro: Box-Supervised 3D Point Cloud Instance Segmentation Using Gaussian Processes as Pseudo Labelers** [Paper] [Webpage]
   Tuan Duc Ngo, Binh-Son Hua, Khoi Nguyen
   ICCV 2023

7. **ISBNet: a 3D Point Cloud Instance Segmentation Network with Instance-aware Sampling and Box-aware Dynamic Convolution** [Paper] [Webpage]
   Tuan Duc Ngo, Binh-Son Hua, Khoi Nguyen
   CVPR 2023

8. **Geodesic-Former: a Geodesic-Guided Few-shot 3D Point Cloud Instance Segmenter** [Paper] [Webpage]
   Tuan Duc Ngo, Khoi Nguyen
   ECCV 2022

Journals

1. **eGAC3D: enhancing depth adaptive convolution and depth estimation for monocular 3D object pose detection** [Paper] [Webpage]
   Tuan Duc Ngo*, Minh-Quan Viet Bui*, Duc Dung Nguyen, Hoang-Anh Pham
   PeerJ Computer Science (Impact Factor: 3.8)

2. **GAC3D: improving monocular 3D object detection with ground-guide model and adaptive convolution** [Paper] [Webpage]
   Minh-Quan Viet Bui*, **Tuan Duc Ngo***, Hoang-Anh Pham, Duc Dung Nguyen
   PeerJ Computer Science (Impact Factor: 3.8)

Pre-prints

1. **DELTAv2: Accelerating Dense 3D Tracking** [Paper] [Webpage]
   Tuan Duc Ngo, Ashkan Mirzaei, Guocheng Qian, Hanwen Liang, Chuang Gan, Evangelos Kalogerakis, Peter Wonka, Chaoyang Wang
   arXiv preprint 2025

Honors & Awards

2025	**Ph.D. Qualifier**, Passed with Distinction	*UMass Amherst*
2025	**CVPR Highlight Paper**,	*CVPR Program Committee*
2023	**CICS Scholarship**,	*UMass Amherst*
2021	**Valedictorian**, graduated with the highest GPA	*VNU*
2017 - 2021	**Academic Encouragement Scholarships**, (8 semesters)	*VNU*
2020	**Honda Award**, Awarded to top 100 undergraduate students in Vietnam	*Honda*

Professional Activities

Talks
- **NECV Workshop** (2024): "DELTA: Dense Efficient Long-range 3D Tracking for any video"
- **ScanNet CVPR Workshop** (2023): "ISBNet: a 3D Point Cloud Instance Segmentation Network"

Conference Reviewer
- **CVPR**: 2024–2026; **ICCV**: 2025; **ECCV**: 2024,2026; **NeurIPS**: 2025; **ICLR**: 2026; **AAAI**: 2025,2026

Journal Reviewer
- IEEE Transactions on Image Processing (**TIP**)

Teaching Assistant
- UMass Amherst: INFO 248 **Introduction to Data Science** (Spring 2026)
- UMass Amherst: COMPSCI 660 **Advanced Information Assurance** (Fall 2025)
- UMass Amherst: COMPSCI 576 **Game Programming** (Fall 2024)
- UMass Amherst: COMPSCI 674 **Intelligent Visual Computing** (Spring 2024)